

18. Ans: C

$$E = \frac{-10 - (-20)}{30 \times 10^{-3}} = 333 \text{ V m}^{-1}$$

The electric force acting on the charge is $(5.0 \times 10^{-6})(333) = 1.67 \times 10^{-3}$ N , and it points

upward. Work done by electric field = $qE \times d = (1.67 \times 10^{-3})(0.012) = 2.0 \times 10^{-5}$ J